
Statistical Foundation

Unit 2

Introduction

In data analysis, understanding the types of variables is fundamental because it determines the kinds of analytical techniques and statistical methods that can be applied.

Two major types:

- Numeric Variables
- Categorical Variables

Numeric Variables

Definition: Numeric variables, sometimes referred to as quantitative variables, represent measurable quantities.

They are expressed with numbers and can be subjected to meaningful mathematical operations such as addition, subtraction, averaging, or finding variance.

Subtypes:

- Continuous Variables
- Discrete Variables

Continuous Variables

Definition: Variables that can take any value within a range, with infinite possible values between two numbers.

Examples:

- Height (e.g., 170.5 cm)
- Weight (e.g., 65.8 kg)
- Temperature (e.g., 36.6°C)
- Time (e.g., 2.75 seconds)

Discrete Variables

Definition: Discrete variables can take only specific, distinct values, typically representing counts.

Examples:

- Number of students in a class (e.g., 30)
- Number of cars in a parking lot (e.g., 15)
- Number of defects on a manufactured item (e.g., 2)

Categorical Variables

Definition: Also called qualitative variables, they represent characteristics, attributes, or labels rather than measurements. They describe qualities and group observations into categories.

Subtypes:

- Nominal Variables
- Ordinal Variables

Nominal Variables

Definition: Classify data into distinct categories where no order or ranking is implied.

Examples:

- Gender (Male, Female, Other)
- Nationality (Nepalese, Indian, American)
- Blood Type (A, B, AB, O)

Ordinal Variables

Definition: Classify data into categories that can be ranked or ordered.

Examples:

- Educational Level (Primary, Secondary, Tertiary)
- Customer Satisfaction (Very Dissatisfied to Very Satisfied)
- Socio-economic Status (Low, Middle, High)

Visual Summary

Type	Subtype	Example	Characteristics
Numeric	Continuous	Temperature	Infinite values in a range
Numeric	Discrete	Number of cars	Specific countable values
Categorical	Nominal	Blood Type	No inherent order
Categorical	Ordinal	Education Level	Ordered categories

Why Does Variable Type Matter?

The type of variable dictates:

- **Choice of Statistical Methods**

- Use mean and standard deviation for continuous numeric data.
- Use mode and frequency for nominal data.

- **Preprocessing Techniques**

- Normalization for continuous variables.
- One-hot encoding for nominal variables.

- **Visualization Methods**

- Histograms for numeric data.
- Bar charts for categorical data.

- **Model Selection**

- Logistic regression handles categorical outcomes.
- Linear regression requires numeric outcomes.

Empirical Distribution

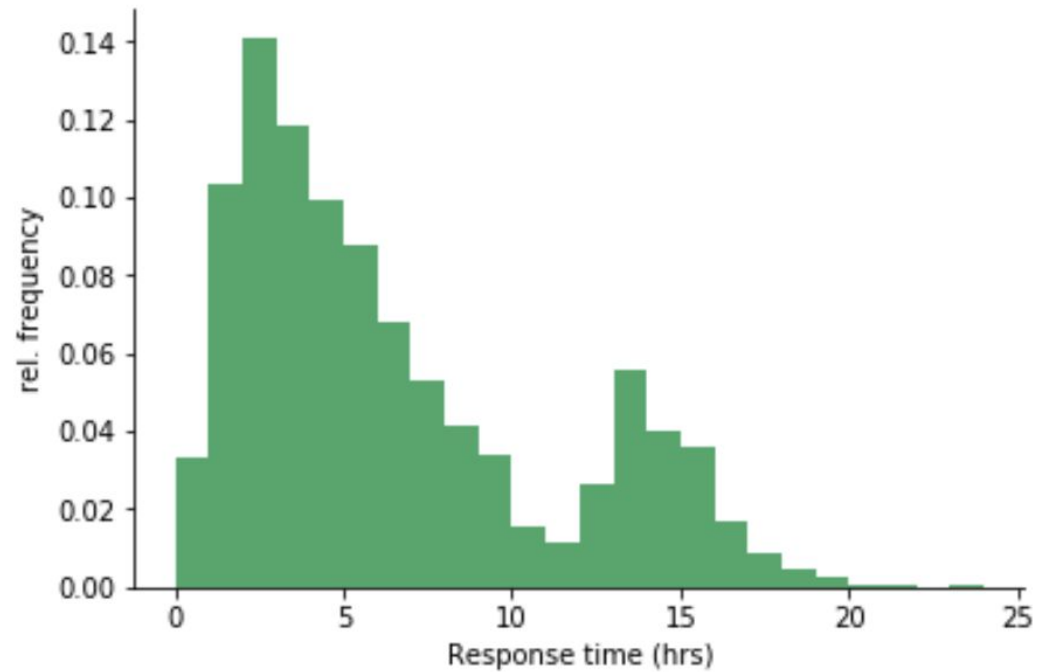
Empirical distributions describe how observed data is distributed, without assuming an underlying theoretical distribution. Understanding empirical distributions is crucial for visualizing and analyzing the behavior of both numeric and categorical data.

Numeric Data

Empirical distribution of numeric data can be visualized and analyzed using various statistical tools and probability models:

- **Histograms:**

- A graphical representation of the frequency distribution of continuous numeric data.
- Data is divided into intervals (bins), and the frequency of data points in each bin is plotted.



Normal Distribution

The **Normal Distribution**, also known as the **Gaussian distribution**, is a probability distribution that is widely used to model real-world numerical data. It has the following characteristics:

- Bell-shaped curve
- Symmetrical around the mean
- Mean (μ), median, and mode are equal
- Defined by two parameters:
 - * Mean (μ): the center of the distribution
 - * Standard deviation (σ): the spread of the data

Mathematical Formula

The probability density function (PDF) of the normal distribution is given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Exponential Distribution

The Exponential Distribution is a continuous probability distribution used to model the time between independent events that happen at a constant average rate.

It is commonly used in fields such as:

- Reliability engineering (time until failure)
- Queueing theory (time between arrivals)
- Survival analysis (time until death or failure)
- Telecommunications (time between packet arrivals)

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

Where:

- λ is the **rate parameter** (average number of events per time unit)
- x is the **time or distance between events**

Power Law

A **Power Law** is a functional relationship where a quantity varies as a power of another. Mathematically, a power-law distribution has the form:

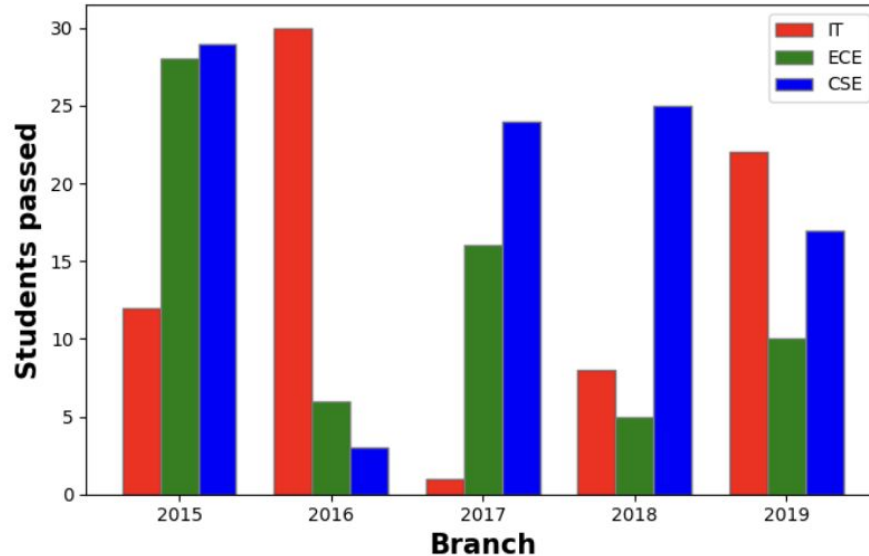
$$P(x) = Cx^{-\alpha}$$

Where:

- x is the variable (e.g., size, frequency)
- $\alpha > 1$ is the **scaling exponent**
- C is the **normalization constant**

- **Bar Plots:**

- Visual representation of frequency or proportion of categories.
- Each bar represents a category, and its height shows the count or relative frequency.



Binomial Distribution

The **Binomial Distribution** is a discrete probability distribution that models the number of **successes** in a fixed number of independent trials, where each trial has only two possible outcomes: **success or failure**.

Conditions for Binomial Distribution

A random experiment follows a Binomial Distribution if:

- The number of trials, n , is fixed.
- Each trial is independent.
- Each trial has only two outcomes: Success (S) or Failure (F).
- The probability of success p remains constant in all trials.

Probability Mass Function (PMF)

The probability of getting exactly k successes in n independent trials is given by:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Zipf's Law

An empirical law: in a large corpus, the frequency of a word is inversely proportional to its rank in the frequency table.

$$f(r) \propto 1 / r^s \quad (\text{with } s \approx 1)$$

$\Rightarrow$ rank $\times$ frequency $\approx$ constant

- **A few words are extremely common** – “the”, “of”, “and” dominate any text
- **Most words are rare** – a long tail of low-frequency terms
- **Why it matters** – guides stop-word removal, indexing & TF-IDF weighting

Zipf's Law – Word Frequencies

Zipf's Law implies that:

- The 2nd most frequent item appears about $\frac{1}{2}$ as often as the most frequent.
- The 3rd most frequent item appears about $\frac{1}{3}$ as often.
- And so on.

This leads to a long-tailed distribution — a few items are very common, while most are rare.

Zipf's Law – Word Frequencies

In a large English corpus:

- **the** is the most common word (rank 1)
- **of** is second (rank 2)
- **and** is third (rank 3)

If **the** appears 10,000 times:

- **of** will appear around 5,000 times
- **and** will appear around 3,333 times

Application of Zipf's law



Linguistics

understanding language structure



Data science

predicting frequencies in ranked data



Information retrieval

keyword weighting (e.g., TF-IDF)



Economics

analyzing inequality & distributions



AI & NLP

language modeling & token-frequency analysis

Correlation

Correlation is a statistical measure that describes the degree to which two or more variables move in relation to each other.

Understanding correlation is essential in various fields, including finance, healthcare, and social sciences, as it helps identify relationships between variables, which can inform decision-making and predictive modeling.

Correlation Analysis

Measures the linear relationship between two numeric variables.

Range: -1 (perfect negative) to +1 (perfect positive).

Key Formulas

Standard Formula:

$$r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

where:

- $\text{cov}(x, y)$: Covariance of x and y
- σ_x, σ_y : Standard deviations of x and y

Alternative Formula:

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Correlation Analysis

Measures the linear relationship between two numeric variables.

Range: -1 (perfect negative) to +1 (perfect positive).

Key Formulas

Standard Formula:

$$r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

where:

- $\text{cov}(x, y)$: Covariance of x and y
- σ_x, σ_y : Standard deviations of x and y

Alternative Formula:

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

When to choose which correlation

Correlation measures the strength & direction of association between two variables. The right technique depends on the variable types involved.



2.3.1

Numeric ↔ Numeric

Pearson correlation (r)

Linear association between two continuous variables. Range -1 to $+1$.



2.3.2

Categorical ↔ Categorical

Cross-tabulation + χ^2

Contingency table, Chi-square test of independence, Cramér's V.



2.3.3

Numeric ↔ Categorical

Point-biserial / ANOVA / η

Compare group means; point-biserial for a binary category.

Assumptions of Pearson Correlation

1 Normality — both variables $\sim$ approximately normal

2 Linearity — relationship is linear (check scatterplot)

3 Homoscedasticity — equal variance across data points

4 No outliers — extreme values can distort r

Interpretation of the correlation coefficient

STRENGTH OF CORRELATION ($|r|$)

$ r \geq 0.7$	Very strong
$0.5 \leq r < 0.7$	Strong
$0.3 \leq r < 0.5$	Moderate
$ r < 0.3$	Weak or none

DIRECTION



Positive ($r > 0$)

as x increases, y increases



Negative ($r < 0$)

as x increases, y decreases

 **Remember:** correlation $\neq$ causation, and r only captures linear relationships — always pair it with a scatterplot.

Find the Pearson Correlation

Age (x)	Salary (y, in \$1,000s)
25	30
32	45
40	55
48	65
55	75

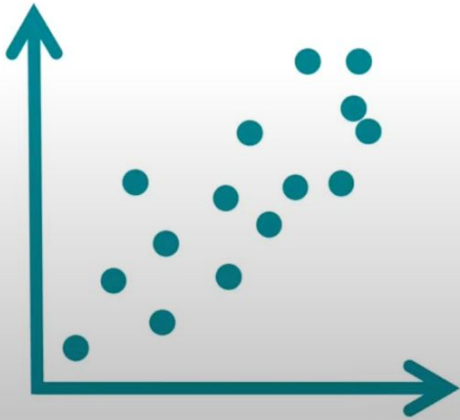
Spearman's Rank

Spearman's rank correlation is a statistical method used to assess the strength and direction of the relationship between two ranked variables

Spearman's rank correlation coefficient, denoted as ρ (rho) or sometimes r_s , measures how well the relationship between two variables can be described using a monotonic function.

It is particularly useful when the data does not meet the assumptions required for Pearson's correlation

Spearman's rank correlation examines the **relationship** between two variables.

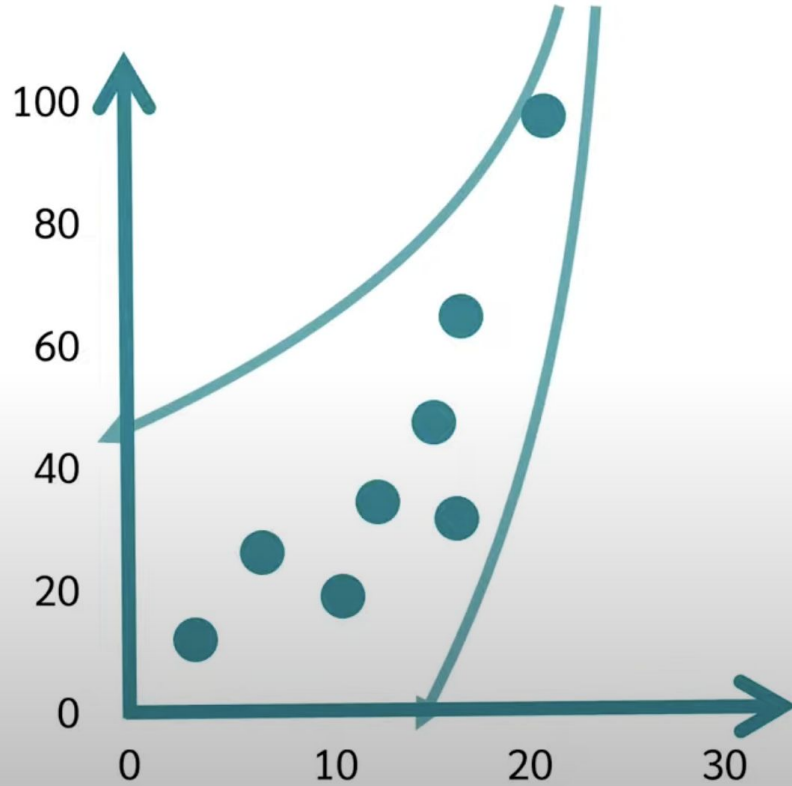


Isn't that exactly what the **Pearson correlation** does?

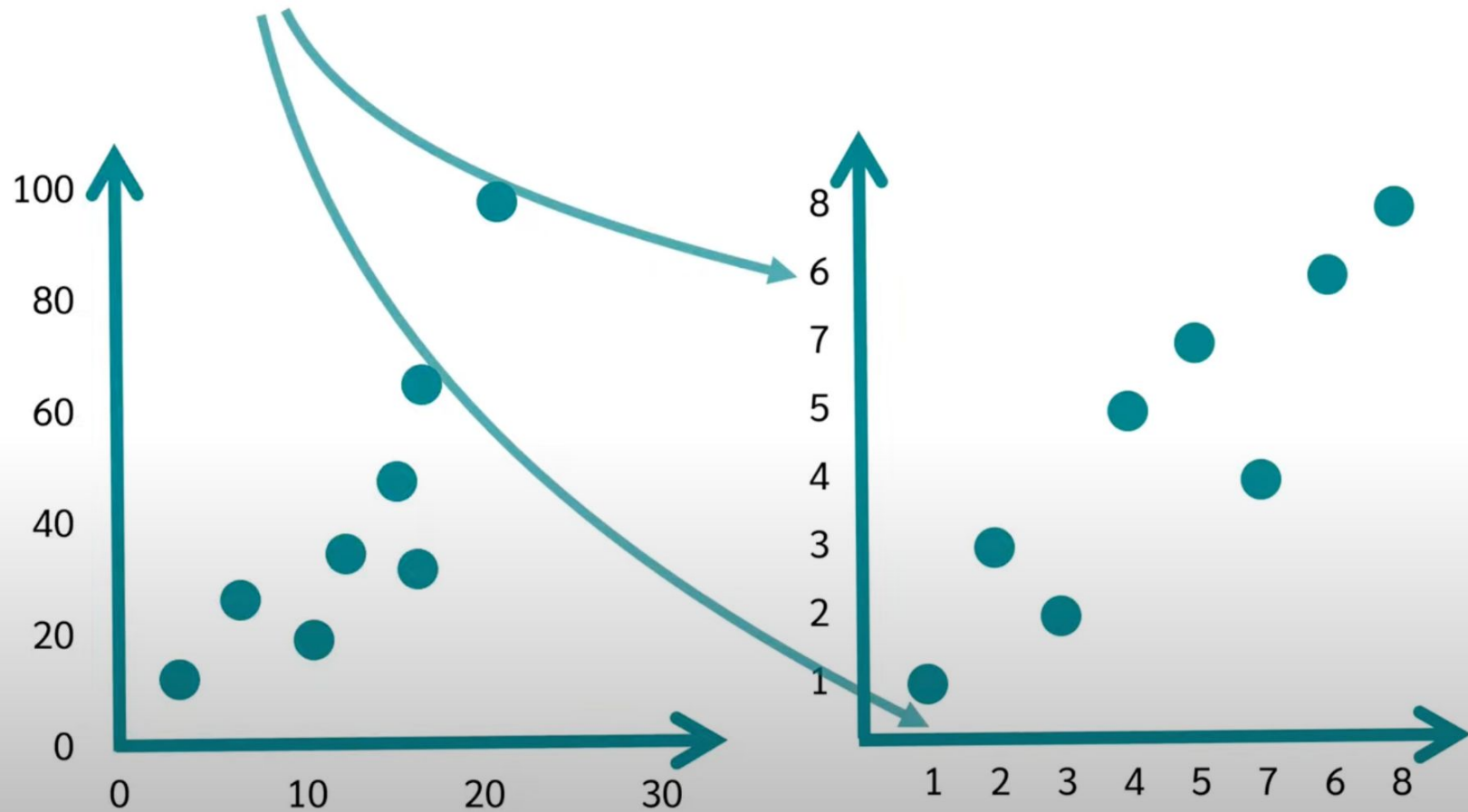


Spearman correlation

does not use the **raw data**,



but the **ranks** of the data.





We measured the **reaction time** of **8 computer players** and asked their **age**.



12

15

17

18

20

21

22

26

When we calculate a **Pearson correlation**, we simply take the two variables **reaction time** and **age**



12
15
17
18
20
21
22
26

14
25
20
35
45
30
60
95

When we calculate a **Pearson correlation**, we simply take the two variables **reaction time** and **age** and calculate the **Pearson correlation coefficient**.


$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$



12
15
17
18
20
21
22
26



14
25
20
35
45
30
60
95

However, we now want to calculate the **Spearman rank correlation**,
so first we assign a **rank** to each
person for **reaction time** and **age**.

		
	12	14
	15	25
	17	20
	18	35
	20	45
	21	30
	22	60
	26	95

The **reaction time** is
already sorted by size.



12 is the smallest value,
so gets **rank 1**



12	1	14
15	2	25
17	3	20
18	4	35
20	5	45
21	6	30
22	7	60
26	8	95

We are now doing the same with **age**.



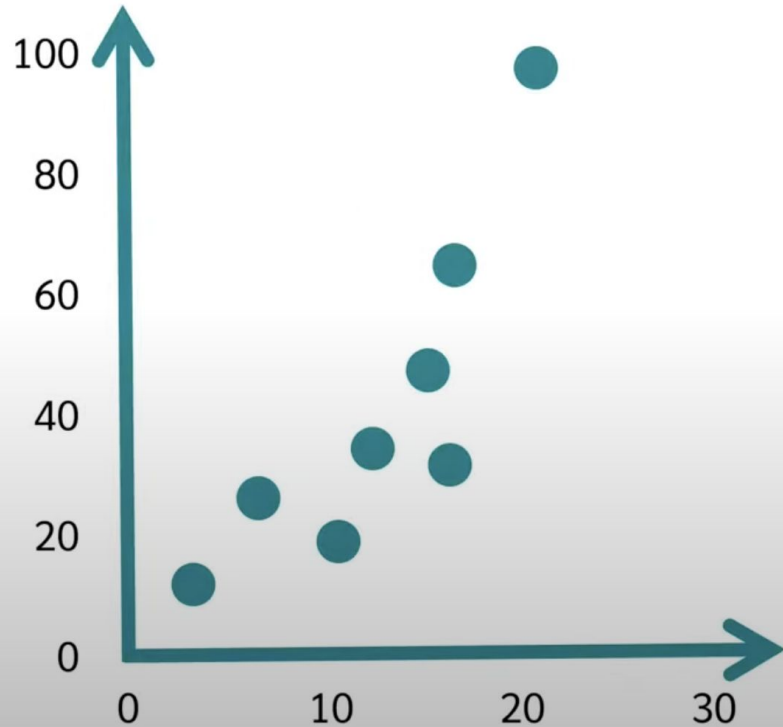
Here we have the **smallest** value,

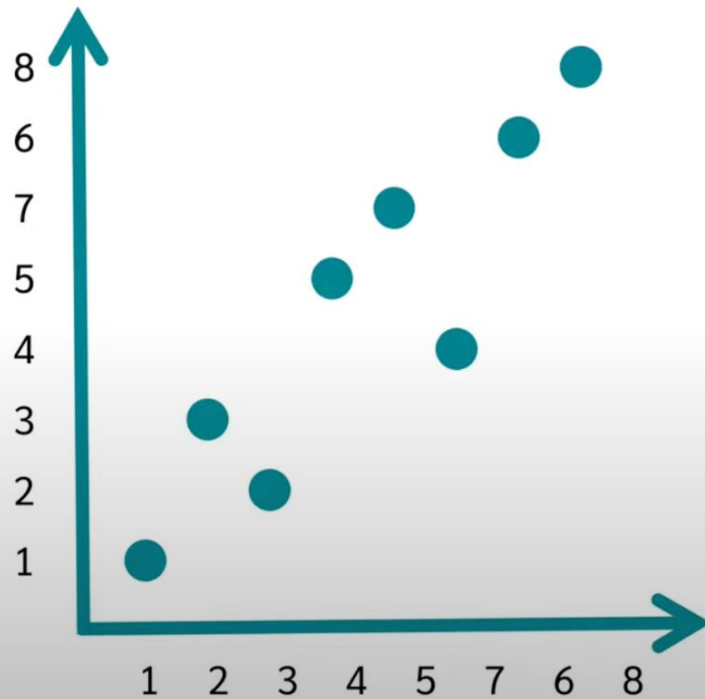
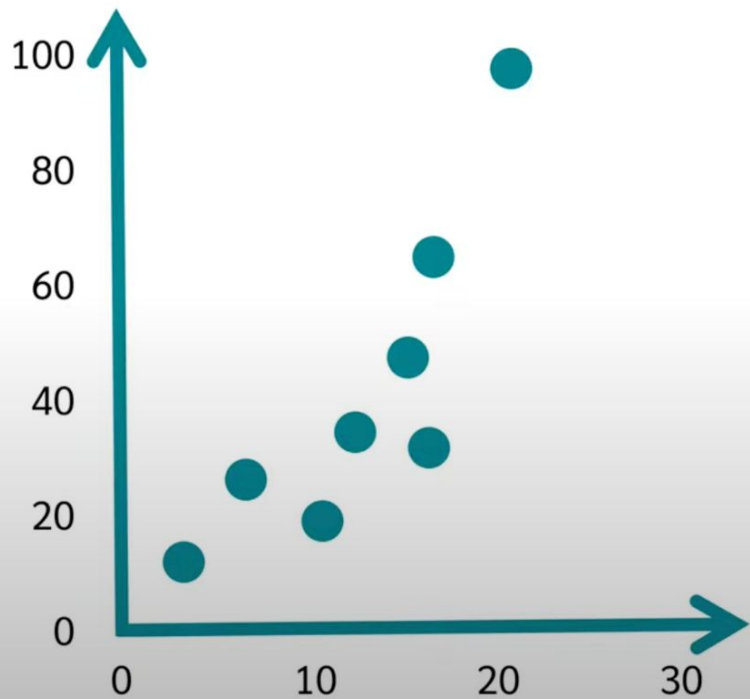


12	1
15	2
17	3
18	4
20	5
21	6
22	7
26	8

14	1
25	3
20	2
35	5
45	6
30	4
60	7
95	8

Let's take a look at this in a **scatter plot**.





If there are **no rank ties**, we can also use this **equation** to calculate the **Spearman correlation**.

$$r_s = 1 - \frac{6 \cdot \sum d_i^2}{n \cdot (n^2 - 1)}$$

n is the **number of cases**

and d is the **difference in ranks** between the two variables.

				d	d ²	
	12	1	14	1	1-1 = 0	0
	15	2	25	3	2-3 = -1	1
	17	3	20	2	3-2 = 1	1
	18	4	35	5	4-5 = -1	1
	20	5	45	6	5-6 = -1	1
	21	6	30	4	7-4 = 2	4
	22	7	60	7	7-7 = 0	0
	26	8	95	8	8-8 = 0	0
						Σ 8

$$r_s = 1 - \frac{6 \cdot \sum d_i^2}{n \cdot (n^2 - 1)}$$

n , which is the number of people, is 8.

If we put everything in, we get a
correlation coefficient of 0.9.

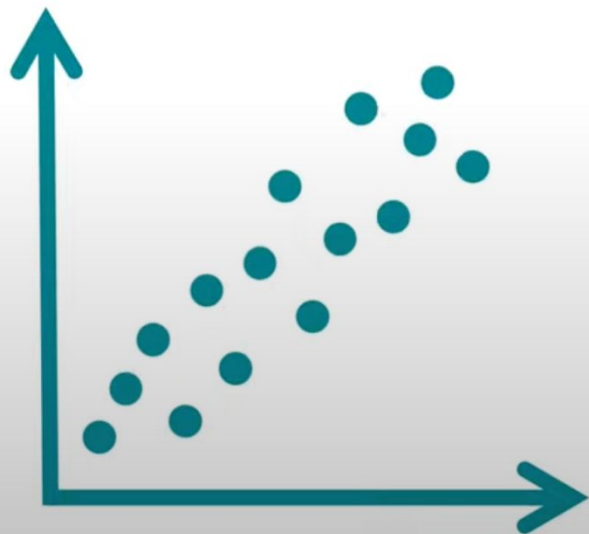

$$r_s = 1 - \frac{6 \cdot \sum d_i^2}{n \cdot (n^2 - 1)} = 1 - \frac{48}{504} = 0.90$$

Just like the **Pearson correlation coefficient r**
the **Spearman correlation coefficient r_s** also varies between **-1** and **1**.



With the help of the **coefficient**, we
can now **determine two things**.

- 1 How **strong** the **correlation** is
- 2 and in **which direction** the **correlation** goes.

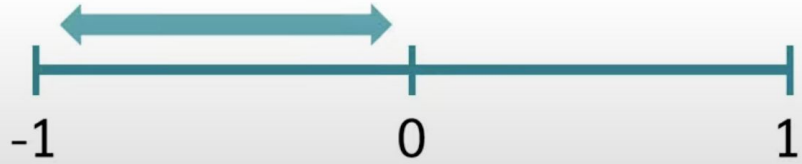


The **strength of the correlation**, can be read in a **table**.

Amount of r	Strength of the correlation
$0.0 < 0.1$	no correlation
$0.1 < 0.3$	low correlation
$0.3 < 0.5$	medium correlation
$0.5 < 0.7$	high correlation
$0.7 < 1$	very high correlation

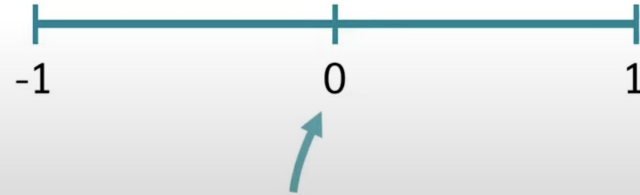
From Kuckartz et al.: Statistik, Eine verständliche Einführung, 2013, p. 213

If we have a coefficient between **-1** and **less than 0**, there is a **negative correlation**,



thus a **negative relationship** between the variables.

If we have a coefficient between **greater than 0** and **1**, there is a **positive correlation**, that is, a **positive relationship** between the two variables.



If the result is **0**, we have **no correlation**.

Often, however, starting from a **sample**,
we want to **test a hypothesis**
about the **population**.



Population



Sample



We calculated the **correlation coefficient** from the **sample data**.
Now we can **test** if the **correlation coefficient** is **significantly different** from **0**.



Thus, the **null hypothesis** is:

The correlation coefficient **$r = 0$**  There is no relationship.

And the **alternative hypothesis** is:

The correlation coefficient **$r \neq 0$**  There is a relationship.

Whether the correlation coefficient is significantly different from zero based on the **sample** collected



can be checked using a **t-test**.

Where r is the **correlation coefficient**

and n is the **sample size**.

$$t = \frac{r \cdot \sqrt{n - 2}}{\sqrt{1 - r^2}}$$

A **p-value** can then be calculated from the test **statistic t**.

$$t = \frac{r \cdot \sqrt{n - 2}}{\sqrt{1 - r^2}}$$

If the **p-value** is less than the specified **significance level**, which is **usually 5%**, then the **null hypothesis** is rejected, otherwise it is not.

Cross Tabulation for Categorical Variables

Cross-tabulation (or crosstab) is a statistical method used to analyze the relationship between two or more categorical variables.

It presents the data in a contingency table, showing the frequency distribution of variables.

Cross Tabulation (Contingency Table)

A crosstab cross-classifies cases by two categorical variables, showing their joint frequency distribution.

Raw data

Gender	Preference	Count
Male	Yes	30
Male	No	20
Female	Yes	40
Female	No	10

Contingency table (crosstab)

	Yes	No	Total
Male	30	20	50
Female	40	10	50
Total	70	30	100

Interpreting the cross tab

1

Frequency distribution

how many cases fall in each combination — e.g., 30 males answered “Yes.”

2

Row %

% within a row — $30/50 = 60\%$ of males said “Yes.”

3

Column %

% within a column — $30/70 = 42.9\%$ of “Yes” answers came from males.

4

Chi-square test

checks whether the variables are independent; $p < 0.05 \Rightarrow$ significant association.

Chi-Square Test of Independence

The χ^2 test evaluates whether two categorical variables are independent or associated in a contingency table.

HYPOTHESES:

H_0 : the variables are independent (no association).

H_1 : the variables are dependent (association exists).

Test Statistic:

$$\chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}$$

O_{ij} = observed frequency in cell (i, j)

E_{ij} = expected frequency under independence

Degree of freedom

$$df = (r - 1)(c - 1)$$

r = number of rows, c = number of columns.

Compare χ^2 to the critical value at chosen significance (e.g., 0.05).

Expected Frequency

$$E_{ij} = \frac{(\text{Row Total}_i) \times (\text{Column Total}_j)}{\text{Grand Total}}$$

Given the cross-tabulation of Gender and Preference and from the table chi-square is 3.84. Declare the hypothesis and determine whether we reject the null hypothesis or failed to reject the null hypothesis.

	Yes	No	Total
Male	30	20	50
Female	40	10	50
Total	70	30	100

Expected Frequencies

For Male/Yes:

$$E_{\text{Male, Yes}} = \frac{50 \times 70}{100} = 35$$

Similarly:

$$E_{\text{Male, No}} = 15, \quad E_{\text{Female, Yes}} = 35, \quad E_{\text{Female, No}} = 15$$

Chi-Square Calculation

$$\chi^2 = \frac{(30 - 35)^2}{35} + \frac{(20 - 15)^2}{15} + \frac{(40 - 35)^2}{35} + \frac{(10 - 15)^2}{15} = 9.52$$

With $df = 1$ and $\alpha = 0.05$, the critical value from χ^2 tables is **3.84**. Since $9.52 > 3.84$, we reject H_0 .

2) Cramer's V (Measure of Strength)

This is used after chi-square to measure how strong the association is.

It is ranged from 0 to 1.

$$V = \sqrt{\frac{\chi^2}{n(k-1)}}$$

where n is the number of sample size

k = smaller of (no. of rows, no. of columns)

i.e. $\min(r-1, c-1)$

Interpretation of V:

Value	Strength
0	No association
0.1–0.3	Weak
0.3–0.5	Moderate
>0.5	Strong

For above question,

$$V = \sqrt{\frac{\chi^2}{n \cdot \min(r - 1, c - 1)}}$$

$$V = \sqrt{\frac{9.52}{100 \times \min(2 - 1, 2 - 1)}} = \sqrt{0.0952} \approx 0.3085$$

Interpretation: Since $V = 0.3085$: moderate association exist between gender and their preferences.

Question: Gender vs Preferred Programming Language

Find cross tabulation and interpretation through Cramer's V rule

Gender	Python	Java	C++	Row Total
Male	25	15	10	50
Female	20	20	10	50
Column Total	45	35	20	100

Assessing Correlation between numeric and categorical

1. Point Biserial Correlation

This method is used to calculate the correlation between numerical and categorical variables.

It is the statistical method that quantifies the relationship between a continuous variable and a binary variable

The calculation of point biserial correlation coefficient involves comparing the means of continuous variables for each category of the binary variable and considering the variability within each category.

Assessing Correlation between numeric and categorical

1. Point Biserial Correlation

The formula for calculating the point-biserial correlation coefficient is

$$r_{pb} = \frac{M_1 - M_0}{S_n} \sqrt{pq}$$

Where,

- M_1 is the mean of the continuous variable for category 1 of the binary variable.
- M_0 is the mean of the continuous variable for category 0 of the binary variable.
- s_n is the standard deviation of the entire population if available.
- p = Proportion of cases in the "1" group. i.e $p=(n_1/n)$
- q = Proportion of cases in the "0" group. i.e. $q=n_0/n$ or $1-p$

Assessing Correlation between numeric and categorical

1. Point Biserial Correlation

The formula for calculating the point-biserial correlation coefficient is

$$r_{pb} = \frac{M_1 - M_0}{S_n} \sqrt{\frac{n_1 n_0}{n(n-1)}}$$

Where,

- M_1 is the mean of the continuous variable for category 1 of the binary variable.
- M_0 is the mean of the continuous variable for category 0 of the binary variable.
- s_n is the standard deviation of the entire population if available.
- n_1, n_0 = Number of observations in groups 1 and 0.
- n = Total sample size ($n_1 + n_0$)

Example: Suppose we have 6 data on households having both numerical and categorical data

Household	Wealth Index	Enrollment
1	170	1
2	175	1
3	180	1
4	160	0
5	165	0
6	162	0

Numerical: Household wealth index

Categorical: Child enrollment in school (1= enrolled, 0= not enrolled)

Calculate Group Means:

- M1: Mean wealth index for enrolled (group 1): $(170 + 175 + 180) / 3 = 175.0$
- M0: Mean wealth index for not enrolled (group 0): $(160 + 165 + 162) / 3 = 162.333$

Calculate the Overall Standard Deviation (sn):

- This is the population standard deviation of the wealth index across all 6 households.
- First, overall mean: $(170 + 175 + 180 + 160 + 165 + 162) / 6 = 168.667$
- Variance: Sum of (each value - mean) ² / 6 = 50.556 i.e (sn) = $\sqrt{50.556} = 7.110$

Calculate Proportions (p and q):

- Total sample size (n): 6, Number in group 1 (n1): 3 i.e. $p = n1/n = 3/6 = 0.5$, Number in group 0 (n0): 3 i.e. $q = n0/n = 3/6 = 0.5$ (or simply $q = 1 - p$)
- $pq = 0.5 \times 0.5 = 0.25$ $\sqrt{pq} = 0.5$

So, correlation coefficient (rpb) = 0.891

This indicates a strong positive association that Higher wealth indices are strongly related to enrollment

Example 2: Relationship between whether a student passed (1=Pass, 0=Fail) and the no. of hours studied

Pass (1/0)	Study Hours
1	12
1	15
1	18
0	5
0	8
0	6

The point-biserial correlation coefficient is calculated as:

$$r_{pb} = \frac{M_1 - M_0}{S_n} \sqrt{\frac{n_1 n_0}{n(n-1)}}$$

$$M1(\text{passed}) = 15$$

$$M2(\text{fail}) = 6.33$$

$$\text{Standard Deviation}(s) = 5.202$$

$$r_{pb} = (15 - 6.333) / 5.202 \times \sqrt{(0.3)} = 0.912$$

There is very strong positive correlation that if a student studies more they will pass

Hypothesis Testing

Introduction

Hypothesis testing is a statistical method used to make inferences or draw conclusions about a population based on sample data.

It involves formulating a hypothesis, collecting data, and then using statistical techniques to determine whether the evidence supports the hypothesis.

Significance Level (α)

The significance level is the threshold for deciding whether to reject the null hypothesis.

Common values are 0.05 (5%) and 0.01 .

It represents the probability of making a Type I error, which occurs when the null hypothesis is incorrectly rejected.

Null Hypothesis (H_0)

The null hypothesis is a statement that there is no effect, no difference, or no relationship between variables in the population. It serves as the default assumption that any observed differences in sample data are due to random chance rather than a true effect.

Examples of Null Hypothesis:

- In a clinical trial, the null hypothesis might state that a new drug has no effect on patients compared to a placebo.
- In a study comparing two teaching methods, the null hypothesis could assert that there is no difference in student performance between the two methods.

The null hypothesis is typically tested using statistical methods, and the outcome of the test will either lead to its rejection or failure to reject.

Alternate Hypothesis (H1)

The alternative hypothesis is a statement that contradicts the null hypothesis. It proposes that there is a significant effect, difference, or relationship between the variables being studied. If the null hypothesis is rejected, the alternative hypothesis is considered to be supported.

Examples of Alternative Hypothesis:

- In the clinical trial mentioned earlier, the alternative hypothesis would state that the new drug does have an effect on patients.
- In the teaching methods study, the alternative hypothesis could claim that one teaching method leads to better student performance than the other.

Hypothesis testing process

1. State the Hypotheses: Clearly define the null and alternative hypotheses.
2. Choose a Significance Level (α)
3. α
4. α): Common choices are 0.05 or 0.01, which represent the probability of rejecting the null hypothesis when it is actually true (Type I error).
5. Collect Data: Gather sample data relevant to the hypotheses.
6. Perform Statistical Test: Use an appropriate statistical test (e.g., t-test, chi-square test) to analyze the data.
7. Make a Decision: Based on the p-value obtained from the test: If

T-test

A t-test is a statistical hypothesis test used to determine whether there is a significant difference between the means of one or two groups. It is particularly useful when the sample sizes are small and the population standard deviation is unknown. The t-test can be categorized into three types:

One-Sample T-Test: Compares the mean of a single group against a known value (e.g., a population mean).

Independent Two-Sample T-Test: Compares the means of two independent groups to see if they are significantly different from each other.

Assumptions of the T-Test:

- The data are continuous and normally distributed.
- The samples are independent (for independent t-tests).
- The variances of the two groups are equal (for independent t-tests).
- The sample data are randomly sampled from the population.

Usage: T-tests are commonly used in various fields, including psychology, medicine, and social sciences, to analyze experimental data and draw conclusions about populations based on sample data.

Chi-square test

The chi-square test is a statistical test used to determine if there is a significant association between categorical variables. It compares the observed frequencies in each category to the frequencies expected if there were no association between the variables.

There are two main types of chi-square tests:

1. Chi-Square Test of Independence: Tests whether two categorical variables are independent of each other. For example, it can be used to determine if gender is independent of voting preference.
2. Chi-Square Goodness of Fit Test: Tests whether the observed distribution of a single categorical variable matches an expected distribution.

Assumptions of the Chi-Square Test:

- The data are categorical.
- The observations are independent.
- The sample size should be sufficiently large, typically at least 5 expected frequencies per category.

Usage: Chi-square tests are widely used in market research, genetics, and social sciences to analyze categorical data and understand relationships between variables.

Importing the libraries

```
import numpy as np
```

```
from scipy import stats
```

Create some sample data

```
# Sample data for traditional method (mean = 75)
```

```
traditional_scores = np.array([70, 75, 80, 65, 78, 72, 74, 76, 77, 79])
```

```
# Sample data for new method (mean > 75)
```

```
new_method_scores = np.array([80, 85, 82, 90, 88, 84, 86, 91, 87, 89])
```

```
# Perform a one-sample t-test
```

```
t_statistic, p_value = stats.ttest_1samp(new_method_scores, 75)
```

```
print(f'Test Statistic: {t_statistic}')
```

```
print(f'P-value: {p_value}')
```

Level of significance

$\alpha = 0.05$

Hypothesis Testing

```
if p_value < alpha:
```

```
    print("Reject the null hypothesis: The new teaching method is effective.")
```

```
else:
```

```
    print("Fail to reject the null hypothesis: No significant effect of the new teaching  
method.")
```

Another data set

```
import numpy as np
```

```
import pandas as pd
```

```
# Set a seed for reproducibility
```

```
np.random.seed(42)
```

```
# Generate daily sales data for each store over 31 days
```

```
dates = pd.date_range(start='1/1/2023', end='1/31/2023')
```

```
# Generate sales data for Store A (mean = 5000, std = 500)
```

```
store_a_sales = np.random.normal(loc=5000, scale=500, size=len(dates))
```

```
# Generate sales data for Store B (mean = 4500, std = 400)
```

```
store_b_sales = np.random.normal(loc=4500, scale=400, size=len(dates))
```

Create the dataset

```
df = pd.DataFrame({  
    'date': np.tile(dates, 2), # Repeat the dates for both stores  
    'store_id': ['A'] * len(dates) + ['B'] * len(dates), # Store identifiers  
    'sales': np.concatenate((store_a_sales, store_b_sales)) # Combine sales  
data  
})
```

Sort the dataset by date

```
df = df.sort_values('date').reset_index(drop=True)
```

```
print(df.head(10)) # Display the first 10 rows of the dataset
```

```
from scipy.stats import ttest_ind
```

```
# Filter data for each store
```

```
store_a_sales = df.loc[df['store_id'] == 'A', 'sales']
```

```
store_b_sales = df.loc[df['store_id'] == 'B', 'sales']
```

```
# Perform two-sample t-test
```

```
t_stat, p_value = ttest_ind(store_a_sales, store_b_sales)
```

```
# Set significance level
```

```
alpha = 0.05
```

```
# Evaluate results
```

```
if p_value < alpha:
```

```
    print("Reject the null hypothesis. There is a significant difference in average  
    daily sales between the two stores.")
```

```
else:
```

```
    print("Fail to reject the null hypothesis. There is no significant difference in  
    average daily sales between the two stores.")
```

```
import pandas as pd
```

```
import numpy as np
```

```
from scipy.stats import chi2_contingency
```

```
# Sample data
```

```
data = {
```

```
    'Names': ['Alice', 'Bob', 'Charlie', 'David', 'Eva', 'Frank', 'Grace', 'Hannah', 'Ian', 'Jack'],
```

```
    'Degree': ['Bachelor', 'Master', 'Bachelor', 'PhD', 'Master', 'Bachelor', 'PhD', 'Master',  
              'Bachelor', 'PhD'],
```

```
    'Years_of_Experience': [2, 5, 3, 10, 7, 1, 8, 4, 6, 9],
```

```
    'Salary': [50000, 70000, 55000, 90000, 75000, 45000, 95000, 80000, 60000, 100000]
```

```
}
```

```
# Create DataFrame
```

```
df = pd.DataFrame(data)
```

```
# Step 1: Categorize Salary
```

```
def categorize_salary(salary):
```

```
    if salary < 60000:
```

```
        return 'Low'
```

```
    elif 60000 <= salary < 80000:
```

```
        return 'Medium'
```

```
    else:
```

```
        return 'High'
```

```
df['Salary_Category'] = df['Salary'].apply(categorize_salary)
```

```
# Step 2: Create Contingency Table
```

```
contingency_table = pd.crosstab(df['Degree'], df['Salary_Category'])
```

```
print("Contingency Table:")
```

```
print(contingency_table)
```

```
# Step 3: Perform Chi-Square Test
```

```
chi2, p, dof, expected = chi2_contingency(contingency_table)
```

```
# Output results
```

```
print(f"\nChi-Square Statistic: {chi2}")
```

```
print(f"P-value: {p}")
```

```
print(f"Degrees of Freedom: {dof}")
```

```
print(f"Expected Frequencies:\n{expected}")
```

```
# Hypothesis Testing
```

```
alpha = 0.05
```

```
if p < alpha:
```

```
    print("Reject the null hypothesis: There is a significant relationship between Degree and Salary  
Category.")
```

```
else:
```

```
    print("Fail to reject the null hypothesis: There is no significant relationship between Degree and  
Salary Category.")
```

```
# Visualizations
```

```
# 1. Bar plot of Degree vs Salary Category
```

```
plt.figure(figsize=(8, 5))
```

```
sns.countplot(data=df, x='Degree', hue='Salary_Category', palette='Set2')
```

```
plt.title('Count of Degrees by Salary Category')
```

```
plt.xlabel('Degree')
```

```
plt.ylabel('Count')
```

```
plt.legend(title='Salary Category')
```

```
plt.show()
```

2. Box plot of Salary by Degree

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(data=df, x='Degree', y='Salary', palette='Set3')
```

```
plt.title('Salary Distribution by Degree')
```

```
plt.xlabel('Degree')
```

```
plt.ylabel('Salary')
```

```
plt.show()
```

3. Histograms of Years of Experience and Salary

```
plt.figure(figsize=(12, 5))
```

Histogram for Years of Experience

```
plt.subplot(1, 2, 1)
```

```
plt.hist(df['Years_of_Experience'], bins=5, color='skyblue',  
edgecolor='black')
```

```
plt.title('Distribution of Years of Experience')
```

```
plt.xlabel('Years of Experience')
```

```
plt.ylabel('Frequency')
```

```
# Histogram for Salary
```

```
plt.subplot(1, 2, 2)
```

```
plt.hist(df['Salary'], bins=5, color='salmon', edgecolor='black')
```

```
plt.title('Distribution of Salary')
```

```
plt.xlabel('Salary')
```

```
plt.ylabel('Frequency')
```

```
plt.tight_layout()
```

```
plt.show()
```

ANOVA Test

ANOVA, or Analysis of Variance, is a statistical method used to determine whether there are significant differences between the means of three or more groups.

It helps researchers understand if the variations in data are due to actual differences between groups or if they can be attributed to random chance.

- One-Way ANOVA: Compares the means of three or more independent groups based on one independent variable. For example, testing the effectiveness of three different diets on weight loss.
- Two-Way ANOVA: Assesses the impact of two independent variables on a dependent variable and can also evaluate interaction effects between the two independent variables. For example, examining the effects of diet type and exercise level on weight loss.

```
import pandas as pd
import numpy as np
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Create the dataset
```

```
data = {
```

```
    'Age': [25, 32, 41, 28, 35, 45, 30, 38, 43],
```

```
    'Income': [50000, 65000, 75000, 55000, 70000, 80000, 60000, 72000,  
78000],
```

```
    'Education': ['Bachelor', 'Master', 'PhD', 'Bachelor', 'Master', 'PhD',  
'Bachelor', 'Master', 'PhD']
```

```
}
```

```
df = pd.DataFrame(data)
```

1. T-test between Bachelor's and Master's

```
bachelor_income = df[df['Education'] == 'Bachelor']['Income']
```

```
master_income = df[df['Education'] == 'Master']['Income']
```

```
t_stat, p_value_ttest = stats.ttest_ind(bachelor_income, master_income)
```

```
print(f"T-test results: t-statistic = {t_stat}, p-value = {p_value_ttest}")
```

2. One-way ANOVA

```
anova_result = ols('Income ~ Education', data=df).fit()
```

```
anova_table = sm.stats.anova_lm(anova_result, typ=2)
```

```
print("\nOne-way ANOVA results:")
```

```
print(anova_table)
```

3. Regression analysis

```
regression_model = ols('Income ~ Age + Education', data=df).fit()
```

```
print("\nRegression analysis results:")
```

```
print(regression_model.summary())
```

```
# Data Visualization
```

```
# Set the style for seaborn
```

```
sns.set(style="whitegrid")
```

```
# 1. Box Plot of Income by Education
```

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(x='Education', y='Income', data=df, palette='Set2')
```

```
plt.title('Income Distribution by Education Level')
```

```
plt.xlabel('Education Level')
```

```
plt.ylabel('Income')
```

```
plt.show()
```

2. Scatter Plot of Age vs Income

```
plt.figure(figsize=(8, 5))
```

```
sns.scatterplot(x='Age', y='Income', hue='Education', data=df, palette='Set1',  
s=100)
```

```
plt.title('Scatter Plot of Age vs Income')
```

```
plt.xlabel('Age')
```

```
plt.ylabel('Income')
```

```
plt.legend(title='Education Level')
```

```
plt.show()
```

3. Histograms for Age and Income

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
```

Histogram for Age

```
sns.histplot(df['Age'], bins=5, kde=True, ax=axes[0], color='skyblue')
```

```
axes[0].set_title('Distribution of Age')
```

```
axes[0].set_xlabel('Age')
```

```
axes[0].set_ylabel('Frequency')
```

```
# Histogram for Income
```

```
sns.histplot(df['Income'], bins=5, kde=True, ax=axes[1], color='salmon')
```

```
axes[1].set_title('Distribution of Income')
```

```
axes[1].set_xlabel('Income')
```

```
axes[1].set_ylabel('Frequency')
```

```
plt.tight_layout()
```

```
plt.show()
```

Numerical

1. P-value

The p-value is the probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis is true.

Formally: $p = P(\text{data} \mid H_0)$

Interpretation

- Small p-value (less than α): Strong evidence against H_0 , reject H_0 .
- Large p-value: Insufficient evidence to reject H_0 .

Misconceptions

- p-value is not the probability that H_0 is true.
- It does not measure the size or importance of an effect.
- It depends on sample size and test design.

T-value (t-test)

The t-value is a test statistic used in the t-test, which compares the means of two groups or a sample mean against a known value when population variance is unknown.

Formula:

For one-sample t-test:
$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

- $\bar{x}$ = sample mean
- μ_0 = hypothesized population mean under H_0
- s = sample standard deviation
- n = sample size

Problem 1: One-Sample t-Test

Question: Given sample data [3.2,3.5,3.7,3.9,4.0] and population mean $\mu_0 = 3.0$, calculate the t-statistic and two-tailed p-value. Assume significance level $\alpha = 0.05$.

Solution:

1. Calculate sample mean ($\bar{x}$):

$$\bar{x} = \frac{\sum x_i}{n} = \frac{3.2 + 3.5 + 3.7 + 3.9 + 4.0}{5} = \frac{19.3}{5} = 3.86$$

2. Calculate sample standard deviation (s):

$$\begin{aligned} s &= \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} \\ &= \sqrt{\frac{(3.2 - 3.86)^2 + (3.5 - 3.86)^2 + (3.7 - 3.86)^2 + (3.9 - 3.86)^2 + (4.0 - 3.86)^2}{4}} \\ &= \sqrt{\frac{0.4356 + 0.1296 + 0.0256 + 0.0016 + 0.0196}{4}} = \sqrt{\frac{0.612}{4}} = \sqrt{0.153} \approx 0.391 \end{aligned}$$

3. Calculate t-statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{3.86 - 3.0}{0.391/\sqrt{5}} = \frac{0.86}{0.1749} \approx 4.914$$

4. Degrees of freedom: $df = n - 1 = 4$

5. Two-tailed p-value (using t-distribution table/calculator):

$$p\text{-value} = 2 \times P(T > 4.914) \approx 0.008$$

Interpretation: Since $p < 0.05$, we reject H_0 at $\alpha = 0.05$.